

A C Govardhan

mail2acgovardhan@gmail.com | linkedin.com/in/acgovardhan | github.com/acgovardhan

RESEARCH INTERESTS

- Blockchain Technology
- AI/ML for Speech & NLP
- IoT Systems

EDUCATION

TKM College of Engineering

B.Tech in Computer Science & Engineering | Honors in Data Science

Sep 2023 – May 2027

Kollam, Kerala

- **CGPA: 9.57 / 10.0**
- Relevant Coursework: Data Structures, Machine Learning, Computer Networks, Database Systems, Operating Systems

Jawahar Navodaya Vidyalaya

Secondary and Higher Secondary

Jul 2015 – Jun 2022

Kollam, Kerala

RESEARCH

Blockchain Technology to Secure Data in IoT Environments

Undergraduate Research Project | TKM College of Engineering | Advisor: Dr. Thushara A

Feb 2026 – Present

- Designing a novel hybrid aggregation protocol combining adaptive batching, Bonawitz-style privacy masking, Merkle-tree anchoring, and BLS signature compression to minimise on-chain footprint for high-volume IoT data streams
- Implementing a multi-criteria adaptive scoring mechanism (freshness, QoS, and urgency) to dynamically size transaction batches, reducing gas costs and improving response time for critical IoT events
- Applying Bonawitz secure aggregation (originally developed for federated learning) to raw IoT sensor data on a public blockchain, enabling aggregate analytics without exposing individual device readings
- Evaluating the proposed protocol against five state-of-the-art baselines (individual transactions, Merkle anchoring, AABF+, BLS-only, and privacy masking) using the IoT-23 dataset across gas cost, latency, bandwidth, and energy metrics

EXPERIENCE

Research Intern

Centre for Development of Imaging Technology (C-DIT)

Sep 2025 – Mar 2026

Thiruvananthapuram, Kerala

- Conducted domain research on the technological barriers faced by Kerala's paddy farming community, identifying low digital literacy, language gaps in existing ASR tools, and lack of dialect-aware voice interfaces as critical unaddressed challenges
- Fine-tuned Wav2Vec2 to build a domain-adapted Malayalam speech-to-text model for paddy farmers, incorporating regional dialects and agricultural vocabulary
- Developing a voice-based data entry and crop analysis tool to aid low-literacy farming communities in Kerala

Full Stack Developer Intern

Deknek 3D (Remote)

Dec 2025 – Mar 2026

- Developed a 3D model animation automation tool in two versions, Unity CLI pipeline and Blender Python API (CLI) to programmatically attach and export animations to arbitrary input models
- Conducted a comprehensive analysis of 20+ 3D platform ecosystems (Sketchfab, Epic Fab etc.), evaluating feature sets, pricing models, API availability, and suitability for web integration
- Built a full-stack AI image generation web app using a Hugging Face diffusion model API
- Designed a student collaboration platform dashboard

Open IoT Student Ambassador

ICFOSS / CDOH

Sep 2024 – Aug 2025

Kerala, India

- Participated in IoT workshops and performed field maintenance of ICFOSS-deployed sensor systems across Kerala
- Repaired and calibrated environmental sensors (rain gauge) at IHRD Engineering College, Kottarakkara, restoring reliable real-time data collection

PROJECTS

SoulSpace - Gamified Mental Wellness Platform

Jan 2025 – Mar 2026

React, Node.js, Express, MongoDB Atlas, JWT, OpenRouter API, OpenStreetMap

- Built a free, full-stack mental wellness platform with clinically validated screenings (PHQ-9, GAD-7, PSS-10, DAT)
- Engineered a gamified wellness space featuring a daily-streak based interface containing virtual pets, doodle pad, and stress-relief mini-games to drive long-term self-care engagement
- Integrated an AI empathetic chatbot via OpenRouter (GPT-4o-mini) grounded in UCPR psychology principles
- Implemented a zero-cost map-based therapist locator using Nominatim (OpenStreetMap), alongside a daily mood tracker

Activity Point Management System

Jul 2025 – Oct 2025

MERN Stack (MongoDB, Express, React, Node.js)

- Digitised KTU's activity point submission and verification workflow via a role-based web platform, replacing manual error-prone paper processes
- Built student submission portal with proof upload and real-time status tracking; developed faculty review dashboard with approval workflow and consolidated report generation

College & Career Guidance System

Jul 2024 – Mar 2025

Java, MySQL (Desktop Application)

- Developed a psychology-based interest assessment engine with course, college, and career recommendation modules using Java Swing and MySQL
- Designed optimised database schemas across Interest Finder, Course Predictor, College Finder, and Career Recommendation engines for fast multi-module retrieval

TECHNICAL SKILLS

Languages: Java, JavaScript (ES6+), Python, C/C++, SQL (MySQL), NoSQL (MongoDB), Solidity (Ethereum)

Frameworks & Web Technologies: React.js, Node.js, Express.js, MERN Stack, Tailwind CSS, Material UI, Astro

AI / ML & Research: PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Hugging Face Transformers

Blockchain & IoT: Ethereum (Solidity), IoT Sensor Systems, Edge Computing

Tools & Platforms: Git, GitHub, VS Code, IntelliJ IDEA, PyCharm, MongoDB Atlas, MySQL Workbench